

ASSIGNMENT 1: WEB TECHNOLOGIES FUNDAMENTALS

Duration: 3-4 hours

Learning Outcomes: HTML/PHP basics, web application architecture, client-server interaction

Tasks:

1. Build basic HTML pages:
 - Create 5 different HTML pages with forms, tables, lists
 - Practice GET and POST methods
 - Use browser developer tools to inspect HTML elements
 - Modify page content using browser console
2. PHP scripting practice:
 - Install XAMPP/WAMP on Windows or LAMP on Linux
 - Create 10 basic PHP scripts (variables, loops, functions)
 - Build simple login form with PHP backend
 - Practice \$_GET and \$_POST variable handling
 - Connect PHP to MySQL database
3. Understand web request/response:
 - Use Burp Suite to intercept HTTP requests
 - Modify GET/POST parameters in real-time
 - Observe cookies and session tokens
 - Analyze HTTP headers (User-Agent, Content-Type, etc.)
4. Source code analysis:
 - View page source on 10 different websites
 - Find hidden fields, comments, and sensitive information
 - Identify web technologies using Wappalyzer
 - Extract form actions and input validation

ASSIGNMENT 2: WEB SERVER HOSTING LAB

Duration: 3-4 hours

Learning Outcomes: Virtual/live server setup, web hosting configuration, server hardening

Tasks:

1. Local virtual server setup:
 - Install Apache web server on Kali Linux
 - Configure virtual hosts for multiple sites
 - Host DVWA on local Apache server
 - Set up PHP and MySQL services
 - Test access from other VMs in your network
2. Cloud hosting experience:
 - Sign up for free tier hosting (000webhost, InfinityFree, or similar)
 - Upload simple PHP application via FTP
 - Configure database through cPanel/phpMyAdmin
 - Access your live hosted application
3. Server configuration exploration:
 - Locate and modify Apache configuration files
 - Enable/disable directory listing
 - Configure .htaccess for URL rewriting
 - Set up error pages (403, 404, 500)
4. Security misconfiguration testing:
 - Test for exposed phpinfo() pages
 - Check for backup files (.bak, .old, .zip)
 - Verify directory traversal protection
 - Test file upload restrictions

ASSIGNMENT 3: WEB APPLICATION PENETRATION TESTING

Duration: 4-5 hours

Learning Outcomes: OWASP Top 10, vulnerability scanning, manual testing techniques

Tasks:

1. DVWA security levels progression:
 - Complete all challenges at Low security level
 - Progress to Medium security level
 - Document what changed between levels
 - Attempt High security level challenges
2. Complete TryHackMe web hacking rooms:
 - OWASP Top 10 room
 - Injection room
 - Web Fundamentals room

- Authentication Bypass room
 - Complete at least 3 web hacking challenges
3. Use automated scanners:
 - Run Nikto against DVWA
 - Use OWASP ZAP active scan
 - Compare results from both scanners
 - Manually verify all findings
 4. Manual testing methodology:
 - Test all input fields for validation
 - Manipulate cookies and session tokens
 - Test file upload functionality
 - Check for information disclosure

ASSIGNMENT 4: SQL INJECTION MASTERY

Duration: 4-5 hours

Learning Outcomes: SQL injection types, exploitation techniques, database enumeration

Tasks:

1. Basic SQL injection in DVWA:
 - Identify SQL injection points
 - Bypass authentication using SQL injection
 - Extract data from database tables
 - Use comment characters to bypass filters
2. Union-based SQL injection:
 - Determine number of columns using ORDER BY
 - Use UNION SELECT to combine results
 - Enumerate database names
 - Extract table names from information_schema
 - Dump usernames and passwords
3. Error-based SQL injection:
 - Trigger SQL errors intentionally
 - Extract database version from errors
 - Use error messages to enumerate structure
 - Practice on DVWA Medium security level
4. Automated SQL injection with SQLMap:
 - Capture request with Burp Suite
 - Run SQLMap against vulnerable parameter
 - Enumerate databases and tables automatically

- Extract specific data using SQLMap options
 - Compare manual vs automated results
5. Blind SQL injection:
- Practice boolean-based blind injection
 - Use time-based blind injection techniques
 - Extract data character by character
 - Automate blind injection with custom scripts

ASSIGNMENT 5: CROSS-SITE SCRIPTING (XSS) EXPLOITATION

Duration: 3-4 hours

Learning Outcomes: XSS types, payload crafting, exploitation impact

Tasks:

1. Reflected XSS practice:
 - Identify reflection points in DVWA
 - Test basic XSS payloads: `<script>alert(1)</script>`
 - Bypass client-side filters
 - Craft working payloads for different contexts
 - Steal cookies using XSS
2. Stored XSS exploitation:
 - Find stored XSS in guestbook/comments
 - Create persistent XSS payloads
 - Demonstrate impact with cookie stealing
 - Use BeEF (Browser Exploitation Framework)
 - Hook victim browsers using XSS
3. DOM-based XSS:
 - Identify client-side JavaScript vulnerabilities
 - Test DOM manipulation payloads
 - Practice on PortSwigger Web Security Academy
 - Understand difference from reflected XSS
4. XSS payload variations:
 - Test 20+ different XSS payloads
 - Bypass filters using encoding (URL, HTML entities)
 - Use event handlers (onerror, onload, onclick)
 - Test polyglot payloads
 - Practice on XSS game challenges
5. XSS impact demonstration:

- Capture keystrokes using XSS
- Redirect users to phishing pages
- Deface web pages temporarily
- Exfiltrate sensitive data